

Exercise Sheet 5

Thursday, January 22, 2026

Note: This exercise lasts for two lab sessions (22.01.2026, 29.01.2026). This exercise is graded and you need to achieve at least 50% of the total points to pass. You can only demonstrate your implementation during the lab sessions on (22.01.2026 and 29.01.2026). You do not need to submit any code, a live demonstration suffices.

5.1 Priority Inheritance Protocol (PIP) (14 points)

In this exercise, PIP should be implemented under the default FreeRTOS fixed-priority scheduler. The central idea on how to implement PIP is outlined in the lecture slides. In essence, when a task with a lower priority blocks a resource that is requested by a higher-priority task, the lower-priority task temporarily inherits the priority of the higher-priority task. After unblocking the resource, the lower-priority task reverts to its original priority.

- Implement PIP in FreeRTOS (10 points)
- Demonstrate that PIP works as expected by using an example where priority inversion occurs without PIP, but is resolved with PIP (4 points)

Hint: You can use your debugging tool to trace the job executions.